Chapter 32

HJM, LMM, and Multi-Curve Term Structure Models †

Section 31.12 put a number on the one-factor defect: a one-factor short-rate model prices a CMS spread option at 56.2 basis points when it is worth 245.0. This chapter is about the class of models built to fix that, and about the price of the fix.

There is a second and less often stated motivation. A short-rate model models a quantity no one quotes. The instantaneous rate $r(t)$ is not on any screen; every market instrument is a function of it, and the model's parameters must be inferred backward through those functions. The market models of this chapter model the rates that are actually quoted, which matters more for calibration than for realism.

The chapter runs from the general framework to the model that is actually implemented. Sections 32.1 to 32.3 are HJM: the whole forward curve at once, the drift no-arbitrage forces on it, and why no one implements it in that form. Sections 32.4 to 32.8 are the LIBOR market model: the measure and the drift that follows, how many factors are needed, and the shapes instantaneous volatility and correlation have to take. Sections 32.9 to 32.13 are practice: calibration, simulation, the model's incompatibility with the swap market model (worth about a basis point of volatility at the money), early exercise, and why sensitivities cost more than prices. Sections 32.14 to 32.16 are what requires this machinery, the smile, and multi-curve after the transition; Sections 32.17 to 32.19 choose among the models and say who runs them.

32.1 HJM: MODEL THE WHOLE FORWARD CURVE

The object being modeled is a family of curves. Section 31.5's $f(0,t)$ was today's forward curve; HJM needs $f(t,T)$, the curve as it stands at any future date t. It is defined as a limit of quoted forward rates. The simply compounded forward rate for $[T, T+\tau]$ seen from t is

$$F(t;T,T+\tau) = \frac{1}{\tau}\left(\frac{P(t,T)}{P(t,T+\tau)} - 1\right)$$

and letting the accrual shrink to nothing gives the instantaneous forward rate,

$$f(t,T) = \lim_{\tau \downarrow 0} F(t;T,T+\tau) = -\frac{\partial \ln P(t,T)}{\partial T} \tag{32.1}$$

Two relations follow, and the chapter uses both throughout. Integrating (32.1) returns the bond price,

$$P(t,T) = \exp\left(-\int_t^T f(t,u)\,du\right) \tag{32.2}$$

so the forward curve and the discount curve carry the same information: either determines the other, and a model of one is a model of both. Evaluating the forward curve at its lower endpoint returns the short rate,

$$r(t) = f(t,t) \tag{32.3}$$

That identity carries the chapter's argument. A short-rate model specifies the single point $f(t,t)$ and derives the rest of the curve from it, which is why the shapes it can produce are constrained. HJM specifies $f(t,T)$ for every T simultaneously, and derives no part of the curve from another.

Heath, Jarrow, and Morton proceeded directly. Instead of a process for one rate, specify a process for every instantaneous forward rate:

$$df(t,T) = m(t,T)\,dt + \sum_{k=1}^{K} s_k(t,T)\,dW_k \tag{32.4}$$

where $m(t,T)$ is the drift of the forward rate for maturity T and $s_k(t,T)$ is its volatility with respect to the kth shock, that is, how far that forward moves when dW_k moves. Both are functions of the date t and the maturity T, so each point on the curve can have its own drift and its own response to each factor, and they may depend on the curve itself.

The HJM processes are written under the risk-neutral measure Q, and each W_k is a Wiener process under it; the market model later changes numeraire, and Section 32.5 says what that does to the drift. No market price of risk appears anywhere in what follows and none is needed: a price is an expectation under Q, and the calibration of Section 32.9 fits Q-parameters to quoted prices directly.

The state is now the entire curve $\{f(t,T) : T \geq t\}$, driven by K independent shocks. Today's curve enters as the initial condition, so the fit that Section 31.4 had to solve for is satisfied by construction. The number of factors is now a modeling choice rather than a consequence. Section 32.6 makes that choice by calculation.

32.2 THE DRIFT IS NOT A CHOICE

Once the volatility functions $s_k(t,T)$ are specified, the drift $m(t,T)$ is determined. No-arbitrage forces

$$m(t,T) = \sum_{k=1}^{K} s_k(t,T) \int_t^T s_k(t,u)\,du \tag{32.5}$$

which is the **HJM drift condition**.

It follows from expressing the bond price in terms of the forward curve and identifying its drift. Under Q a zero-coupon bond earns the short rate, so the bond's drift is determined before any modeling choice is made, and the drift of the forward curve must be the one consistent with it.

Take one factor for the derivation and drop the subscript; the K-factor result follows at the end. Take logs in (32.2) and name the result,

$$Y(t,T) \;\equiv\; \ln P(t,T) \;=\; -\int_t^T f(t,u)\,du$$

Differentiation produces two terms. The integrand evolves, contributing $-\int_t^T df(t,u)\,du$; and the lower limit advances, evaluating the integrand at that limit and contributing $+f(t,t)\,dt$. That second term is where the short rate enters, by (32.3):

$$dY = r(t)\,dt - \int_t^T \big[m(t,u)\,dt + s(t,u)\,dW\big]du = \big[r(t) - M(t,T)\big]dt - S(t,T)\,dW$$

where

$$M(t,T) = \int_t^T m(t,u)\,du, \qquad S(t,T) = \int_t^T s(t,u)\,du$$

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are the drift and the volatility of the forward curve accumulated across the bond's remaining life. That accumulation is the essential feature: a bond responds to the entire strip of forward rates it spans, not to any single one of them.

Apply Itô's lemma to $P = e^Y$ to get

$$\frac{dP}{P} = \left[r(t) - M(t,T) + \frac{1}{2} S(t,T)^2 \right] dt - S(t,T)\, dW$$

A zero-coupon bond is a traded asset and (32.4) is written under Q, so the bond's expected return is the short rate, which leads to

$$M(t,T) = \frac{1}{2} S(t,T)^2$$

That equation is the entire content of the drift condition: the accumulated drift is not a free input, but is determined by the accumulated volatility.

M and S are both integrals over $[t,T]$, so differentiating in T gives $\partial M/\partial T = m(t,T)$ and $\partial S/\partial T = s(t,T)$, with the chain rule on the right:

$$m(t,T) = S(t,T)\, s(t,T) = s(t,T) \int_t^T s(t,u)\, du$$

which is (32.5) with one factor. With K shocks, S becomes a vector and the variance term becomes $\frac{1}{2} \sum_k S_k^2$; the shocks are independent, so no cross terms appear, each factor contributes the one-factor result, and the contributions add.

The derivation used only that bond prices are traded and that $f(t,t) = r(t)$. No equilibrium argument and no utility function appear, which is why the condition binds every arbitrage-free term-structure model rather than a family of them. Three consequences follow, and the third is the one that shapes practice.

The model has exactly one free input. Choose the volatility structure and everything else follows. It is the analog of Chapter 13's result that a derivative's price does not depend on the underlying's expected return: drift is not an independent degree of freedom once you insist on no arbitrage.

Every arbitrage-free term-structure model is an HJM model. Two of Chapter 31's Gaussian models are (32.4) under particular choices of $s(t,T)$:

- *Ho–Lee*: $s(t,T) = \sigma$, the same volatility at every maturity, so a shock shifts the whole curve in parallel;
- *Hull–White*: $s(t,T) = \sigma e^{-a(T-t)}$, a volatility decaying with the forward rate's distance, so a shock moves the short end more than the long.

Chapter 31 was therefore working inside HJM, with a volatility structure chosen for tractability rather than for fit. The third consequence is that the one input is a function of two variables, which is why the framework is hard to implement.

32.3 WHY HJM IS NOT IMPLEMENTED DIRECTLY

Two obstacles stand in the way, and they are not of the same severity. The first removes Chapter 31's lattices; the second only requires the model to be restated in other variables.

The first is that the process is generally non-Markov: unless $s(u,T)$ is a finite sum of terms, each a function of u times one of T, no finite set of numbers summarizes the accumulated shocks $\int_0^t s(u,T)\, dW(u)$ for every maturity at once, so the future depends on the path. A non-Markov process cannot be put on a recombining tree, which removes Chapter 31's trees and leaves only Monte Carlo—and Monte Carlo, by Section 20.11, is awkward for early exercise.

There is an important exception, and it explains Chapter 31. If $s(t,T) = \sigma e^{-a(T-t)}$ (volatility decaying exponentially in time to maturity), the model *is* Markov in a single state

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variable, and that state variable is the short rate. Hull–White is the HJM model that happens to be Markov in the short rate alone (Ho–Lee is its $a = 0$ case), which is why it can be put on a one-dimensional tree while a general HJM model cannot be put on any. The tractability that made Chapter 31 useful was never an accident of the short-rate formulation; it was a restriction on the volatility structure.

The exponential case is worth following through, because it is the one that reduces to a single state variable. With $s(t,T) = \sigma e^{-a(T-t)}$ the integral gives $S(t,T) = \sigma B(t,T)$, Hull–White's own function, and integrating the forward-rate dynamics leaves

$$f(t,T) = f(0,T) + \phi(t,T) + e^{-a(T-t)}x(t), \qquad x(t) = \int_0^t \sigma e^{-a(t-u)}\,dW(u)$$

with ϕ deterministic. The entire curve is a deterministic function of the single scalar $x(t)$, an **Ornstein–Uhlenbeck process** (a Gaussian process pulled back at rate a toward a fixed level, here zero, of the kind Vasicek's model makes of the short rate), and setting $T = t$ shows x is the short rate up to a deterministic shift.

The step that collapses it is a factorization: $e^{-a(T-u)} = e^{-a(T-t)}e^{-a(t-u)}$ separates the volatility's dependence on calendar time from its dependence on maturity. That is the precise sense in which Hull–White, with Ho–Lee as its limit, is the simplest Markov HJM model; a second exponential buys a second state variable, and the pattern continues.

The exception is a family, not one model. An HJM volatility built from separable terms,

$$s(t,T) = \sum_i g_i(t)h_i(T),$$

with the g_i allowed to depend on the current state, is Markov in a small state vector; one deterministic term is Hull–White with time-dependent parameters, and the linear Gaussian Markov (LGM) and Cheyette models are this family. They are used when a Bermudan needs a skew or a second factor but not a full LMM.

The second obstacle concerns calibration, and it is not that instantaneous forward rates are untraded. The short rate is not traded either, and Chapter 31 put it on a trinomial tree and priced Bermudan swaptions there; an unobservable state variable is the normal case in this subject, not an objection to it. The difficulty is one of dimension. HJM's free input $s_k(t,T)$ is a function of two variables (a volatility for every pair of calendar date and maturity), while the market quotes a finite set of volatilities, on rates that accrue over three or six months rather than instantaneously. No market instrument pays off on a single $f(t,T)$, so there is no quote to read any one of those volatilities from. A model written in terms of $f(t,T)$ must therefore be discretized along the maturity axis before it can touch a quote, and that discretization is a modeling choice the framework does not supply, which is to say the modeling happens in the discretization rather than in (32.4).

The market model removes the second obstacle and accepts the first: it is no more Markov than HJM and is simulated too, but it works with rates that are quoted, over the accrual periods they are quoted over.

32.4 THE LIBOR MARKET MODEL

Fix the tenor structure $T_0 = 0 < T_1 < \cdots < T_N$ of an actual instrument (the reset dates of a cap or a swap) and model the forward rates $F_k(t)$ for the periods $[T_k, T_{k+1}]$ directly, each fixing at the start of its own period, T_k. Taking $T_0 = 0$ has one consequence used throughout: F_0 fixes today, so it is known rather than modeled, carries no caplet, and still supplies the discount factor for the first period. The caplets of a cap on this structure are therefore on F_1 through F_{N-1}, and the one on F_k has k whole periods of life. The convention costs no generality, because the grid and the instrument are separate things: a structure that starts later (a cap beginning in five

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years, or the swap underlying a swaption) keeps this same grid from today and simply carries no caplets on the forwards before it starts. Those forwards are still modeled, since the discount factors to the later payment dates run through them. Assume each forward is lognormal:

$$\frac{dF_k(t)}{F_k(t)} = \mu_k(t)\,dt + \sigma_k(t)\,dW_k \tag{32.6}$$

with $dW_i\,dW_j = \rho_{ij}\,dt$.

The assumption makes the model consistent with the market's own quoting convention. Section 29.3 priced a caplet with Black's model, which assumes exactly this lognormality. So the LMM reproduces the market's caplet prices by construction rather than by calibration: the model is built on the convention instead of being fitted to it.

Before the LMM, the market quoted caps in a formula no arbitrage-free term-structure model contained. Afterward, the quoting formula was a special case of the model. The name is now an anachronism (the rates are SOFR, €STR and EURIBOR forwards, not LIBOR), but the construction is unchanged, and Section 32.16 says why. Caps and swaptions have been quoted mostly in normal volatility since rates went negative (Section 29.8), and a normal or shifted LMM reproduces those caplet prices by construction in the same way.

32.5 WHICH MEASURE, AND THE DRIFT THAT FOLLOWS

Under the T_{k+1}-forward measure $Q^{T_{k+1}}$ (numeraire $P(t,T_{k+1})$), the forward F_k is a martingale and $\mu_k = 0$. But different forwards are martingales under different measures, and a simulation needs all of them under one.

Pick a single numeraire and the drifts follow. Under the terminal measure (numeraire $P(t,T_N)$),

$$\mu_k(t) = -\sigma_k(t) \sum_{j=k+1}^{N-1} \frac{\rho_{kj}\,\tau_j\,F_j(t)\,\sigma_j(t)}{1+\tau_j F_j(t)} \tag{32.7}$$

The derivation is a change of numeraire performed once per accrual period, and one object does all of the work. For each j write

$$\Lambda_j(t) \;=\; \frac{P(t,T_{j+1})}{P(t,T_j)}$$

for the ratio of two consecutive numeraires. By the definition of F_j that ratio is a function of a single forward rate and of nothing else,

$$\Lambda_j(t) \;=\; \frac{1}{1+\tau_j F_j(t)}$$

It carries the letter this book uses for a Radon–Nikodym derivative because it is one up to a constant:

$$\left.\frac{dQ^{T_{j+1}}}{dQ^{T_j}}\right|_{\mathcal{F}_t} \;=\; \frac{P(0,T_j)}{P(0,T_{j+1})}\,\Lambda_j(t)$$

and a constant multiple leaves a volatility unchanged.

Since F_j has volatility σ_j, applying Itô's lemma to $\Lambda_j = 1/(1+\tau_j F_j)$ gives

$$\frac{d\Lambda_j(t)}{\Lambda_j(t)} \;=\; \mu_{\Lambda_j}(t)\,dt \;-\; \lambda_j(t)\,dW_j, \qquad \lambda_j(t) \;=\; \frac{\tau_j F_j(t)\,\sigma_j(t)}{1+\tau_j F_j(t)}$$

The drift μ_{Λ_j} is never used and is not evaluated. The volatility of Λ_j is λ_j: F_j's own volatility, damped by the factor $\tau_j F_j/(1+\tau_j F_j)$, the damping that makes the drift small, as the end of this section shows. The minus sign in the display is no part of that volatility, since $-W_j$ is a

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Wiener process as much as W_j is; it records that Λ_j falls when F_j rises, and that is where the minus sign on the covariance below comes from.

Section 28.9 derived this as Girsanov's theorem: a change of numeraire adds to a process's drift its covariance with the numeraire ratio, whatever that drift already is, and leaves the volatility unchanged. For consecutive bond numeraires the ratio is Λ_j, and because Λ_j moves against F_j, a forward F_k of volatility σ_k and correlation ρ_{kj} with F_j has covariance $-\rho_{kj}\,\sigma_k\,\lambda_j$ with it. Stepping from Q^{T_j} to $Q^{T_{j+1}}$ therefore adds exactly that to μ_k.

The walk goes one numeraire at a time, from the forward's own measure to the terminal one. Under $Q^{T_{k+1}}$ the forward F_k is a martingale and $\mu_k = 0$; this is the measure the caplet on F_k is priced in. Take one step, to $Q^{T_{k+2}}$. The numeraire ratio for that step is Λ_{k+1}, and with $dW_k\,dW_{k+1} = \rho_{k,k+1}\,dt$ from (32.6) the rule gives

$$\mu_k(t) \;=\; -\,\sigma_k(t)\,\rho_{k,k+1}\,\lambda_{k+1}(t)$$

The correlation enters for one reason: the covariance the rule asks for is between the diffusion of F_k and that of Λ_{k+1}, and F_{k+1} is the only forward rate Λ_{k+1} contains.

Take the next step, to $Q^{T_{k+3}}$. The ratio is now Λ_{k+2}, the identical calculation runs with $j = k+2$, and a second term $-\sigma_k\rho_{k,k+2}\lambda_{k+2}$ is added to the drift F_k already carries. In general the step from Q^{T_j} to $Q^{T_{j+1}}$ uses Λ_j and contributes $-\sigma_k\rho_{kj}\lambda_j$.

The contributions add rather than interact, because consecutive ratios multiply:

$$\frac{P(t,T_N)}{P(t,T_{k+1})} \;=\; \prod_{j=k+1}^{N-1} \Lambda_j(t)$$

The logarithm of a product is the sum of the logarithms, so the diffusion terms add (not the volatilities, since the factors are driven by different Wiener processes), and the whole change from $Q^{T_{k+1}}$ to Q^{T_N} carries $-\sum_{j=k+1}^{N-1}\lambda_j\,dW_j$. Walking all $N-k-1$ steps and collecting,

$$\mu_k(t) \;=\; -\,\sigma_k(t)\sum_{j=k+1}^{N-1} \rho_{kj}\,\lambda_j(t)$$

which is (32.7). Only forward rates longer than F_k appear, because the walk crosses exactly Λ_{k+1} through Λ_{N-1} and no others.

Simulating under the **spot measure** instead (whose numeraire is a rolling investment, rolled at each reset date into the bond maturing at the next) gives sums over $j \le k$, with the opposite sign. Neither choice is more correct. The terminal measure leaves the last forward driftless; under the spot measure none is, but each sums only over the forwards up to its own. Implementations choose by where the drift, and its discretization error, does least harm. Three observations matter for implementation.

The drift is state-dependent, since F_j appears on the right. That is what makes the simulation of Section 32.10 nontrivial and is the source of its discretization error.

The drift is a no-arbitrage consequence, not a parameter. It is the same statement as (32.5), in the discretized language.

It is also small. At a 3% rate with annual accrual, $\tau F/(1+\tau F) \approx 0.029$, so each term in the sum is roughly three percent of a product of volatilities. That smallness has a consequence worth stating in advance of Section 32.10: the predictor–corrector scheme exists because a plain Euler step freezes this state-dependent drift across the interval, and switching to the scheme moves Example 32.4's swaption volatility by 0.15 basis points, while raising the step count from four to two hundred moves it by about 6. The discretization error is not where the state dependence would put it.

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32.6 HOW MANY FACTORS

The whole motivation was Section 31.12's factor count, so it should be settled by calculation rather than convention.

Example 32.1 The curve's factor structure

No rates are simulated and no data are used: consider thirty one-year forward rates ($T_i = 1, \ldots, 30$), assume their correlation is $\rho_{ij} = e^{-\beta|T_i - T_j|}$, and compute the eigenvalues of that matrix. Every figure in the table is computed from that matrix at an assumed β; none is a market estimate. The cumulative share of variance explained is:

β	1 factor	2	3	4	5
0.02	82.8%	92.5%	95.3%	96.6%	97.4%
0.05	64.8%	82.2%	88.6%	91.7%	93.5%
0.10	46.6%	68.1%	78.3%	83.9%	87.3%
0.20	28.9%	49.0%	61.9%	70.2%	75.9%

This is **principal component analysis**: the eigenvectors are the factors, the eigenvalues their shares of variance. The eigenvectors carry the reading Litterman and Scheinkman found in bond returns and every major market has reproduced since: a roughly parallel shift (**level**), one sign change (**slope**), two (**curvature**).

Take β of 0.05 to 0.10 as an illustrative range for major currencies, which puts the first factor at 47–65% and the first three at 78–89%. Historical studies of forward rates show the same three shapes (level, slope, and curvature), with three factors explaining less than the 98–99% found for yields, by an amount that depends on the data and on how the curve is interpolated.

The form is an assumption, not a finding. What recommends it is that it is a valid correlation matrix for every $\beta \geq 0$, so no admissibility condition has to be checked, and that it decays monotonically in tenor distance, the robust qualitative feature of measured correlation. In practice only β is estimated; the shape is imposed.

Neither the volatility structure nor the drift can move these percentages, since the matrix is a correlation matrix. The number of forwards can: at $\beta = 0.05$ the first factor explains 85.5% of ten annual forwards, 64.8% of thirty, 57.5% of forty.

The form also sends ρ_{ij} to zero far out, where measured correlation flattens to a floor; the standard repair is $\rho_{ij} = \rho_\infty + (1 - \rho_\infty)e^{-\beta|T_i - T_j|}$. A floor is itself a common factor, so it lifts every entry: at $\rho_\infty = 0.3$ and $\beta = 0.05$ one factor explains 75.2% and three 92.0%. The table is conservative, and 92% is still not 99%.

Against one factor the table is decisive: it leaves a third to a half of the variance unexplained, and Example 31.11 priced what that omission does to a spread option.

Against the often-repeated "three factors explain 99% of the curve," it is a corrective. That figure comes from principal components of changes in *yields* (about 98% in Litterman and Scheinkman), and a yield averages the forwards out to its maturity, so neighboring yields move together far more than forwards do: averaged into yields, this same matrix gives three factors 98.7–99.4%. On forward-rate correlations of the assumed form, which are what a spread option depends on, three factors explain 78–89%—and the residual is exactly the relative movement the option is written on.

The working answer is three factors for most books and two for many, and more for a book holding spread options: on this form three keep only about half the volatility of a ten-year-less-two-year swap spread, and seven nearly all. The marginal factor is cheap: under simulation its cost grows linearly, where on a lattice it would grow exponentially with the dimension.

32.7 INSTANTANEOUS VOLATILITY AND ITS SHAPE

Section 29.5 distinguished flat volatility (one number per cap, the number the market quotes) from spot volatility (one number per caplet, the caplet volatilities that together reproduce the

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cap's price). The LMM needs a third object: the *instantaneous* volatility $\sigma_k(t)$, a function of calendar time for each forward.

Spot volatility σ_k^{Black} and $\sigma_k(t)$ are tied by requiring that total variance match:

$$\sigma_k^{\text{Black},2}\, T_k = \int_0^{T_k} \sigma_k(t)^2\, dt \tag{32.8}$$

The market determines the integral and not the integrand, so the shape of $\sigma_k(t)$ is a modeling choice with no caplet price to discipline it. The standard choice is Rebonato's parametrization, in which volatility depends only on the time left to maturity, $T_k - t$, and not on where the calendar happens to stand. A specification with that property is **time-homogeneous**, and the term recurs below:

$$\sigma_k(t) = \big[a_R + b_R(T_k - t)\big]e^{-c_R(T_k - t)} + d_R \tag{32.9}$$

Four parameters produce a humped shape, which is what the data show (Rebonato 2004): a forward rate two years out is more volatile than one thirty years out and also more volatile than one settling next week. Four parameters cannot match every caplet, so implementations scale the form by a constant per forward, $\sigma_k(t) = \Phi_k\, g(T_k - t)$, where g is the function of $T_k - t$ on the right of (32.9), each Φ_k set by (32.8); how far the Φ_k stray from one measures the departure from time homogeneity.

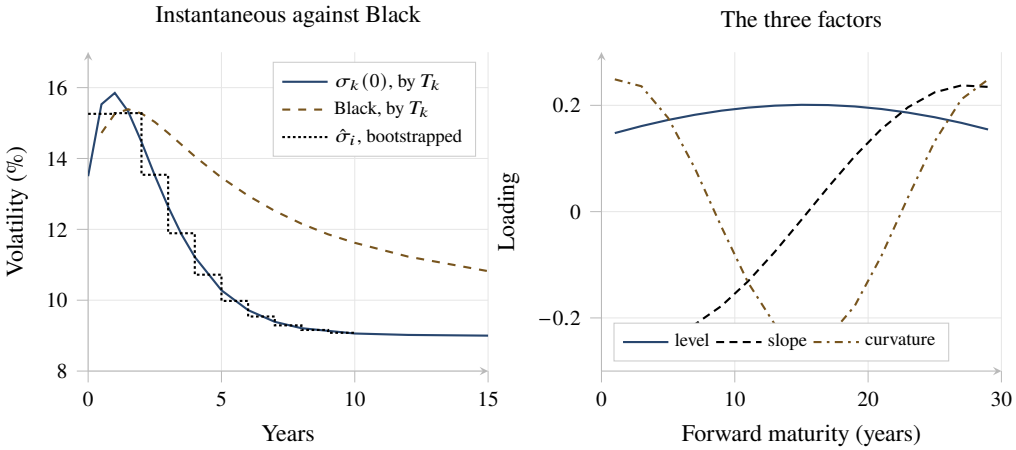


Figure 32.1: Left: equation (32.9) at $a_R = 0.045$, $b_R = 0.10$, $c_R = 0.75$, $d_R = 0.09$, the Black volatilities it implies through (32.8), and (dotted) the piecewise-constant levels bootstrapped from them. Right: the first three eigenvectors of Example 32.1 at $\beta = 0.05$. Illustrative parameters, not a calibration.

Figure 32.1's left panel shows why the distinction matters. The two smooth curves are today's, and both are indexed by the caplet's own maturity: at T_k the solid curve is that forward's instantaneous volatility now, the dashed one its Black volatility. The two can share an axis only because of time homogeneity. As t runs from 0 to T_k the argument $T_k - t$ sweeps $[0, T_k]$, which is exactly the interval the solid curve covers, so the instantaneous volatilities a caplet meets over its life are the values the solid curve takes to the left of its own maturity, and the dashed curve is the running root-mean-square of the solid one.

The instantaneous volatility peaks at 15.87% about ten and a half months out (at $1/c_R - a_R/b_R$) and falls to $d_R = 9\%$ in the tail; the Black volatilities it implies peak at 15.39% and are still at 10.8% fifteen years out, where the panel ends. Both curves tend to d_R in the end (a

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running root-mean-square of a function approaching 9% approaches 9%), but the dashed one takes far longer to get there, reaching 9.95% only by thirty years. A desk that reads a quoted Black volatility as though it were the instantaneous volatility of that forward has the wrong number by more than three volatility points around five years, and still by nearly two at fifteen. The right panel shows the eigenvectors of Example 32.1: the first is positive throughout, the second changes sign once near fifteen years, the third twice.

A second specification is in wide use, and it is worth knowing because it makes the freedom in (32.8) concrete. Instead of a smooth function, take $\sigma_k(t)$ **piecewise constant** (one level per accrual period) and time-homogeneous. Write $\hat{\sigma}_i$ for the level in the i-th period counting backward from the reset, so that $\hat{\sigma}_1$ is the level over the last period before the rate fixes. Then (32.8) is a sum rather than an integral,

$$\sigma_k^{\text{Black},2}\, T_k \;=\; \sum_{m=1}^{k} \hat{\sigma}_{k-m+1}^2\, \tau_{m-1} \tag{32.10}$$

counting forward: period m runs for τ_{m-1} and carries the level with $k-m+1$ periods still to run. When the accruals are all of one length τ, the right-hand side is $\tau \sum_{i=1}^{k} \hat{\sigma}_i^2$, and differencing consecutive caplets inverts it:

$$\hat{\sigma}_k^2\, \tau \;=\; \sigma_k^{\text{Black},2}\, T_k \;-\; \sigma_{k-1}^{\text{Black},2}\, T_{k-1} \tag{32.11}$$

Each caplet in turn supplies one more level. No optimizer is involved, and the fit to the caplet strip is exact by construction. That is the attraction: the instantaneous volatilities come off the quoted strip with a subtraction.

Only equal accruals reduce to that subtraction, and the condition is not a technicality: on a strip alternating one-year and six-month periods with levels falling from 16% to 11%, using (32.11) anyway (with $\tau = \tau_0$) gets $\hat{\sigma}_2$ wrong by a third and $\hat{\sigma}_6$ by more than half. The general case costs one more line rather than an optimizer. The level with k periods to run appears in (32.10) for caplet k exactly once, in the $m = 1$ term, so each caplet still brings exactly one new unknown and the bootstrap still runs one caplet at a time; what gets subtracted is the remainder of the sum rather than the previous caplet's total variance:

$$\hat{\sigma}_k^2\, \tau_0 \;=\; \sigma_k^{\text{Black},2}\, T_k \;-\; \sum_{m=2}^{k} \hat{\sigma}_{k-m+1}^2\, \tau_{m-1}$$

The cost is the mirror image. There are as many free levels as caplets, so nothing smooths the result, and nothing prevents (32.11) from returning a negative $\hat{\sigma}_k^2$, which it does whenever the quoted strip falls steeply enough that a later caplet's total variance falls short of an earlier one's. The parametric form cannot fail that way, and on its own cannot fit exactly either; the Φ_k multipliers are what make it exact.

The two specifications look very different as functions even when they agree on every caplet. On the strip implied by (32.9) at the parameters above, the bootstrap gives

Periods to reset	1	2	3	4	5	6	7	8	9	10
$\hat{\sigma}_i$, instantaneous	15.26	15.28	13.54	11.89	10.72	9.98	9.54	9.29	9.16	9.08
Caplet Black vol	15.26	15.27	14.72	14.06	13.46	12.95	12.52	12.16	11.86	11.62

in percent. The instantaneous levels fall to 9.1% where the caplet volatilities they reproduce fall only to 11.6%, because each caplet volatility averages the instantaneous one over the whole life. Figure 32.1's left panel draws the smooth form, these levels and the Black curve they share.

The reason to prefer a time-homogeneous form is not fit but stability: the volatility structure such a model implies for one year from now looks like today's. A model that is not time-homogeneous predicts that the market's volatility surface will change shape by tomorrow—not

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because anything happened, but because the calendar advanced—and a desk hedging vega on such a model is hedging an artifact. Section 21.3 made the same criticism of local volatility's forward smile, and it is the same defect. Both forms above are time-homogeneous; Example 32.2 prices the choice between them on three products built to see the difference.

Example 32.2 What the shape is worth: three nonstandard caps

Two specifications that price every caplet identically must differ somewhere, and the products that find the difference are the ones whose payoff depends on the path of the forward-rate vector rather than on one rate at one date. Three standard ones follow, each a cap whose strike or exercise rule is not fixed in advance:

- a **ratchet cap** resets each strike to the rate just observed plus a spread, $K_{k+1} = R_k + s$;
- a **sticky cap** resets it to the previous *capped* rate plus a spread, $K_{k+1} = \min(R_k, K_k) + s$, so the strike follows the rate down but rises by at most the spread each period;
- a **flexi cap** is an ordinary cap on which the first n caplets that finish in the money pay.

All three are sold to floating-rate borrowers who want protection for less premium than a vanilla cap; the dealer hedges them with vanilla caps and keeps the shape exposure measured below.

Take annual periods on the grid of Section 32.4, so that the cap's ten caplets are the ones on F_1 through F_{10} and the terminal measure is $Q^{T_{11}}$. Put the forwards flat at 5%, take the correlation of Example 32.1 at $\beta = 0.05$, a spread of 25 basis points, vanilla and flexi strikes of 5%, ratchet and sticky caps starting from $K_1 = F_0 + s = 5.25\%$, and a flexi cap limited to five exercises. Calibrate both specifications to the same caplet strip (the parametric form to (32.9)'s parameters, the piecewise-constant form by the bootstrap (32.11) from the volatilities that form implies), so that every vanilla caplet price agrees to eight decimals. Then simulate both on the same draws:

	Parametric	Piecewise	Difference	Std. err.
Ratchet cap	1.8098	1.8064	−0.19%	0.02%
Sticky cap	3.6035	3.5992	−0.12%	0.02%
Flexi cap	2.7835	2.7812	−0.08%	0.03%
Vanilla cap	4.1499	4.1501	0.01%	0.01%

Values are per 100 of notional, from 48,000 antithetic pairs at sixteen steps a year, with each step's covariance integrated exactly and both specifications on the same draws.

The vanilla cap is a control rather than a result. Both specifications reproduce its price (a Black value of 4.1414, inside the simulated level's standard error of 0.013) by construction, so any difference there is numerical, and 0.01% against a standard error of 0.01% shows that the experiment is sound. The ratchet's −0.19% is therefore shape and not noise, at more than nine standard errors; the sticky cap's −0.12% likewise; the flexi cap's −0.08% is under three of its own standard errors: the flexi cap barely sees the shape.

Hold each σ_k at its value at the start of the step instead, at four steps a year, and the vanilla cap reads +0.31% (discretization bias in the smooth form, which a step volatility on the reset grid escapes) while the ratchet reads −1.55%, eight times the table's figure, and nothing inside the ratchet's own number reveals it. Example 32.4 meets the same lesson from the other side: a quantity computed as a difference between two models inherits both models' discretization error, and the only defense is an instrument whose answer you already know.

A ratchet at 0.19% of premium is well inside most desks' bid–ask on a ten-year structure, but the exposure is one-directional: the smooth form is dearer on both the ratchet and the sticky cap. The reason is not where the volatility sits: each bootstrapped level is the root-mean-square of the smooth curve over its own period, so the two forms put the same variance into every period. It is how neighboring forwards move together within a period. The step function moves them in lockstep, so it correlates successive fixings slightly more, and a ratchet or sticky cap, whose strike is set by the previous fixing, is an option on the change from one fixing to the next. A desk that calibrates one way and hedges against a counterparty who calibrates the other keeps that difference

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as an unexplained mark.

32.8 CORRELATION: INSTANTANEOUS AND TERMINAL

Two distinct quantities carry the same name and are routinely confused. **Instantaneous correlation** ρ_{ij} describes how the two forwards move together over an instant. **Terminal correlation** describes how their values at some horizon are related, and that is what a spread option pays on.

Example 32.3 Do the two differ enough to matter?

With the Rebonato form (32.9) at $a_R = 0.045$, $b_R = 0.10$, $c_R = 0.75$, $d_R = 0.09$, terminal correlation over a horizon H is

$$\frac{\rho_{ij}\int_0^H \sigma_i\sigma_j\,dt}{\sqrt{\int_0^H \sigma_i^2\,dt\int_0^H \sigma_j^2\,dt}}$$

Set the instantaneous correlation to 1 and see what the volatility shapes alone produce:

T_i	T_j	Horizon	Terminal corr. at $\rho = 1$
2	3	2y	0.9968
2	5	2y	0.9970
2	10	2y	0.9993
5	10	5y	0.9936
10	20	5y	0.9994

The shapes alone give at most 0.6% of decorrelation, which looks ignorable and is not. A spread's variance is $V_i + V_j - 2\rho\sqrt{V_iV_j}$ with $V_i = \int_0^H \sigma_i^2\,dt$, and the subtraction nearly cancels at high ρ, so the surviving error is amplified: treating the instantaneous ρ as the terminal one understates the 5y/10y spread volatility by 1.25% at $\rho = 0.85$ and 4.01% at $\rho = 0.99$. An at-the-money premium moves in proportion to its volatility, so it is understated by the same percentages, which at high correlation reach the size of a swaption's bid–ask.

So in the LMM the correlation you calibrate is close to the correlation you get, which makes the correlation matrix the object to spend calibration effort on and the model's spread-option prices interpretable. It does not make the shape correction droppable: a multiplier of 0.994 is not a multiplier of one once the cancellation amplifies it.

The contrast with Chapter 31 sharpens too. A one-factor short-rate model has $\rho_{ij} = 1$ by construction and no offsetting shape effect either, so its terminal correlations are essentially 1 as well. The defect of Section 31.12 cannot be repaired with clever volatility structures; it takes more factors.

32.9 CALIBRATION IN PRACTICE

Caplet volatilities come first and are exact. Equation (32.8) determines the total variance of each forward from the market's caplet quotes, and the LMM reproduces them by construction. No optimization is involved.

The volatility shape is then fitted subject to that constraint. The four parameters of (32.9) are chosen to fit swaption volatilities (which, unlike caplets, depend on the shape and on correlation), with the Φ_k re-solved at each trial so the total variances stay at their caplet values; the correlation enters at a first estimate and is revisited below.

Correlation is fitted last and is the hard part. It can be estimated historically, parametrized (the $e^{-\beta|T_i-T_j|}$ form of Example 32.1 is standard), or implied from spread options where those are liquid. Historical and implied correlations have tended to differ, and by more than the volatility analogs of Section 19.10. That should be expected. A **correlation risk premium**

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(a persistent gap between the correlation priced into options and the correlation the rates go on to show) arises for the same reason as a volatility risk premium, but far less trading can close it: rate volatility is bought and sold directly, through a deep cap and swaption market, whereas the correlation between two points on the curve trades only through CMS spread options themselves and through relative-value positions in swaptions against caps, both of which carry other exposures and leave the dealers on one side (Section 32.14). A gap between historical and implied correlation can therefore persist with no one able to arbitrage it away.

Then the correlation matrix is reduced to K factors, which changes it. A rank-K approximation to a full correlation matrix does not reproduce the original entries, and Example 32.1's table is exactly the statement of how much is lost.

LMM calibration is therefore one exact step, one well-posed fit, and one genuinely underdetermined estimate.

32.10 SIMULATION

The LMM has no lattice, so everything is Monte Carlo, and the drift (32.7) is state-dependent. That state dependence rules out the exact scheme of Section 20.7, which is exact only because a geometric Brownian motion's drift is deterministic. What is used instead is an Euler step taken on the logarithm. Itô's lemma turns (32.6) into

$$d \ln F_k = \left(\mu_k(t) - \frac{1}{2} \sigma_k(t)^2 \right) dt + \sigma_k(t)\, dW_k$$

and stepping that over $[t, t + \Delta t]$ gives

$$F_k(t + \Delta t) \;\approx\; F_k(t) \exp\left[\left(\mu_k(t) - \frac{1}{2} \sigma_k(t)^2 \right) \Delta t + \sigma_k(t)\, \varepsilon \sqrt{\Delta t} \right]$$

It is an approximation in both coefficients: each is held at the value it has when the interval opens, and neither is constant across it, σ_k least of all, since Section 32.7 gives it a hump.

The two are not on the same footing, though. Because σ_k is deterministic, its approximation is removable: use the variance the model actually accumulates over the step, $\int_t^{t+\Delta t} \sigma_k(u)^2\, du$, in place of $\sigma_k(t)^2 \Delta t$, and the diffusion becomes exact (with the covariances $\int \sigma_j \sigma_k\, du$ treated the same way). The drift admits no such treatment, because μ_k depends on the forwards themselves and is not known until the step has been taken. One approximation can be integrated away and the other has to be corrected for, which is why a predictor–corrector exists for the drift and nothing analogous is needed for the diffusion.

Stepping the logarithm rather than F_k itself keeps the forward positive, since it is an exponential, so the negative forwards a direct Euler step can produce never arise. With the variance integrated exactly, the frozen drift would be the only error left anywhere, and it is what the predictor–corrector removes. The comparison below keeps the step as displayed, with both coefficients frozen, so that it can measure the two errors side by side: the gap between the two schemes is the drift's error, and the movement as the step count rises is mostly the volatility's.

The standard concern is that freezing the drift introduces bias, and that the predictor–corrector of Section 20.7 removes it. Measuring it needs a product the model cannot price in closed form. A cap will not do: the LMM reproduces every caplet exactly by construction, so simulating one tests the simulation against an answer already in hand. A swaption is the simplest instrument whose value depends on the forwards jointly (on their correlations as well as their volatilities), so the model has no formula for it and it has to be simulated. It is also the trade Section 32.11 needs, which is why the same one appears there.

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Example 32.4 Where the simulation error actually is

A one-year at-the-money option on a five-year swap, five annual forwards flat at 3%, instantaneous correlation 0.85, Rebonato volatility as in Example 32.3. Four hundred thousand antithetic pairs were used.

Each simulated price is inverted through Black's swaption formula of Section 29.6 to give the Black volatility the third column reports. That assumes nothing about the swap rate: Black's value is monotone in volatility, so the inversion is a change of units, and it states the error in basis points of volatility rather than in currency. The last column then measures each figure against Rebonato's approximation, which Section 32.11 sets out; the approximation serves only as a fixed yardstick here, so the table's subject is the movement down the column, not the distance across to it.

Steps	Scheme	Implied vol	vs. approximation
4	Euler (frozen drift)	12.52%	−6.2 bp
4	Predictor–corrector	12.52%	−6.3 bp
12	Euler	12.53%	−5.2 bp
12	Predictor–corrector	12.53%	−5.3 bp
50	Euler	12.58%	−0.2 bp
50	Predictor–corrector	12.58%	−0.2 bp
200	Euler	12.58%	−0.1 bp
200	Predictor–corrector	12.58%	−0.1 bp

The Monte Carlo standard error is 1.8 basis points of volatility on each level, and rows at different step counts do not share draws, so a difference between two of them carries about 2.5. The difference between the two schemes at one step count is another matter: it shares its draws, and is measured to a thousandth of a basis point. Two findings follow, and the second contradicts the usual advice.

Discretization bias is real and it goes away quickly. Four steps a year leaves the answer about 6 basis points low. That is about two and a half standard errors of the difference, but the shortfall is not noise: it is what holding σ_k at its value at the start of each step does across the hump, and computing that effect directly, with no simulation, gives 7.7 basis points at four steps and 0.6 at fifty. Fifty steps closes the gap to inside the Monte Carlo error, and two hundred does not improve on fifty.

The predictor–corrector makes no measurable difference here. On shared draws the two schemes differ by 0.15 basis points at four steps a year and by 0.003 at two hundred—far smaller than the discretization bias, at every step count.

The reason is Section 32.5's third observation. Every term of (32.7) carries a product $\sigma_k\sigma_j$, so the drift is second order in volatility where the diffusion is first order, and the factor $\tau F/(1+\tau F) \approx 0.029$ damps it further. Over this option's one-year life the largest drift moves $\ln F$ by 0.0019, against a diffusion standard deviation of 0.153. Correcting a term that small cannot move the answer, and the residual bias is coming from the frozen volatility (the error that integrating the variance removes) rather than from the frozen drift.

The drift correction earns its cost when the drift is large (high rates, and many forwards between the simulated rate and the numeraire) and not otherwise. The accrual convention matters less than it looks: halving τ halves each term of (32.7) but doubles the number of terms over the same stretch of curve. A desk running a five-forward book at 3% that pays for a predictor–corrector has bought nothing, while one running a thirty-forward book in a high-rate currency on coarse steps needs it. This is Section 20.7's rule: a predictor–corrector is for a drift that depends on the state and is not small beside the diffusion, and which case applies has to be measured on the problem at hand.

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32.11 THE SWAP MARKET MODEL, AND WHAT THE INCOMPATIBILITY COSTS

Section 29.6 priced swaptions with Black's model applied to the forward swap rate, which assumes the *swap rate* is lognormal. Section 32.4 assumes the *forward rates* are lognormal.

Behind the first assumption is a model built exactly as the LMM is. Section 28.8 showed the forward swap rate $S(t)$ to be a traded asset divided by the annuity, and therefore a martingale under the annuity measure Q^A. Give it a deterministic volatility and

$$\frac{dS(t)}{S(t)} = \sigma_S(t)\, dW^A$$

which is the **swap market model**. A swaption is then priced by Black's formula exactly, in the same sense and for the same reason that the LMM prices a caplet exactly: each model takes the rate its own market quotes, makes that rate lognormal under the measure in which it is already a martingale, and so reproduces the quoting convention by construction rather than by calibration.

That is the one-rate case, and it is all the argument below needs. The swap market model in full carries a family of co-terminal swap rates (swaps starting on different dates and all ending on the same one) under a single annuity measure, with drift conditions derived exactly as Section 32.5's were and with swap rates in place of forwards; it is Jamshidian's, and this chapter states it without developing it, because the derivation would repeat Section 32.5 with the letters changed while the LMM is the model more commonly implemented. Its natural application is the Bermudan swaption, whose exercise dates are precisely a co-terminal strip, so there the model's state variables are the very swap rates the holder can exercise into. That is the other route to the problem Section 32.12 takes in LMM terms.

Both cannot be exactly true. A swap rate is a weighted sum of forwards with stochastic weights, and a weighted sum of lognormals is not lognormal. This is a theorem, and it means the market's two standard quoting conventions are mutually inconsistent.

That statement usually ends the discussion. The useful question is how large the inconsistency is, and it has a number.

Example 32.5 Measuring the inconsistency

Rebonato's approximation freezes the weights with which the swap rate averages the forwards, which makes the swaption's Black variance over the option's life T a weighted average of the forwards' covariances:

$$\sigma_S^2 T \;\approx\; \sum_{i,j} w_i w_j\, \rho_{ij} \int_0^T \sigma_i(t)\, \sigma_j(t)\, dt, \qquad w_k = \frac{\tau_k P(0,T_{k+1})\, F_k}{A(0)\, S(0)}$$

where σ_S is the swaption's Black volatility and A is the annuity; on a flat curve, as here, these equal the elasticities $(\partial S/\partial F_k)(F_k/S)$. For the swap of Example 32.4 these are 0.2120, 0.2058, 0.1998, 0.1940, 0.1884, summing to 1.0000.

The approximation gives a Black volatility of **12.5818%**. Simulating it gives **12.58%**, 0.1 basis point away.

Now change the trade, since a short option on a short swap is where an approximation has the least to do. A five-year at-the-money option on a ten-year swap, ten annual forwards flat at 6%, instantaneous correlation 0.70:

	1y into 5y at 3%	**5y into 10y at 6%**
Rebonato approximation	12.5818%	9.0048%
Simulation	12.58%	8.99%
Error	0.1 bp	1.3 bp

The longer trade is worse, as it should be, and still negligible. With the swap and an option on

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each forward as control variates, whose values are known exactly, the simulation's error falls below 0.1 basis point: the approximation is 0.1 basis point high on the first trade and 1.3 on the second, with the same sign under two numeraires, and neither is visible against a bid–ask of about half a volatility point.

That the weights sum to one is an identity, not a normalization. Each term $\tau_k P(t, T_{k+1}) F_k$ equals $P(t, T_k) - P(t, T_{k+1})$, so the sum telescopes to $A(t)S(t)$ and $\sum_k w_k = 1$ on any curve. This is why the check above is informative: a weight vector that misses one signals an error in the discount-factor recursion rather than a modeling choice.

The approximation looks cruder than it is. The weights do move: more with more forwards, with longer expiration, and with larger τF, all three of which grow between the two columns. But the displacements are driven by the same zero-mean shocks as the rates, so across paths they average out to first order, and at the money the error enters at second order in them rather than at first. Because the weights keep summing to one, a weight that drifts up is offset by others drifting down, which keeps even a single path's first-order effect proportional only to how much the forwards' covariances with the swap rate differ.

So the swap/forward incompatibility is a theorem, but at the money it is worth at most about a basis point across the range tested, far inside the bid–ask; away from the money the sum of lognormals has a skew of its own, which Section 32.15's models absorb. A desk may calibrate to caps and price swaptions with Rebonato's formula and not think about it again—which is what desks in fact do, and this is the calculation that says they are entitled to.

The general lesson is one this book has drawn twice before, and this is the first time it comes out the other way. Section 21.3 found local volatility's forward-smile error structural and large; Section 31.12 found the one-factor correlation error structural and a factor of four. Here the structural error is real as mathematics and negligible as a price—which is the most comfortable of the three answers and the rarest.

32.12 EARLY EXERCISE

A Bermudan swaption needs a continuation value at every exercise date, and Monte Carlo runs forward while the exercise decision needs backward induction. Section 20.11's Longstaff–Schwartz regression is the standard answer: regress realized continuation values on state variables at each exercise date, and use the fitted function as the exercise rule. Three practical points follow, since this is where implementations go wrong.

The regression basis must span the state. With a high-dimensional forward-rate vector, the natural choices are the swap rate and the annuity, sometimes with a second swap rate to capture the curve's shape. Choosing too small a basis produces a suboptimal exercise rule and therefore a downward-biased price, which looks like conservatism and is not.

The estimate is biased in both directions and the biases are separable. Using the same paths to fit the rule and to price gives an upward foresight bias; using a suboptimal rule gives a downward bias. The standard discipline is to fit on one set of paths and price on another, which removes the first and leaves the second, so the reported price is, up to sampling error, a genuine lower bound. Upper bounds via the dual formulation (which values exercise with perfect foresight, less a martingale chosen to charge for that foresight) exist and are expensive.

The whole apparatus is also slower than a tree by orders of magnitude. Example 31.8's Bermudan takes milliseconds on a trinomial tree; the same product in a three-factor LMM with regression takes seconds to minutes. For a book of thousands of Bermudans revalued nightly with sensitivities, that difference decides the model, which is the reason short-rate models still run the callable book.

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32.13 SENSITIVITIES, AND WHY THEY COST MORE THAN PRICES

A price is one number; a book needs its derivatives with respect to every calibration input, and in an LMM there are many. The obvious method is **bump-and-revalue**: move one input by a small amount, revalue, and take the difference against the unperturbed price. It scales badly. With M curve points and V volatility points, a full sensitivity run is $M + V + 1$ simulations (the base, plus one for each input). At fifty curve points and thirty volatility points, that is eighty-one Monte Carlo runs for one product, so an LMM book's overnight risk run, not its pricing, is the binding computational constraint.

Common random numbers are not optional here. If the bumped and base simulations use different draws, the difference of two noisy estimates carries the noise of both, which can exceed the bump's whole effect, so the delta comes out smaller than its own standard error. Section 20.10 made the point for equity Greeks, where sharing the draws improves an estimate; in an LMM risk run it decides whether there is an estimate at all, and Example 32.6 measures the gap.

Example 32.6 Why the draws must be shared

A swaption prices at 100.0000 with a Monte Carlo standard error of 0.0500 at 100,000 paths, and a one basis point bump to the curve moves it by 0.0480.

With independent draws the difference of the two runs has standard error $0.0500\sqrt{2} = 0.0707$:

$$\frac{\text{signal}}{\text{noise}} = \frac{0.0480}{0.0707} = 0.68$$

The delta is smaller than the uncertainty in the delta—the run has produced almost no information, and a risk report built this way is reporting sampling noise with a sign.

With common random numbers the two runs differ only through the bump, most of the sampling error cancels, and a standard error near 0.0004 is routine:

$$\frac{\text{signal}}{\text{noise}} = \frac{0.0480}{0.0004} = 120$$

The cost of not sharing the draws is not a factor of two; it is a factor of 31,250 in paths. Matching that precision with independent draws requires

$$\left(\frac{0.0707}{0.0004}\right)^2 \times 100{,}000 \approx \textbf{3.1 billion paths}$$

per bump, and there are eighty bumps. One line of implementation—seed the bumped run identically—buys what no amount of hardware can.

The same arithmetic explains why the input count is what kills the method rather than the path count. Adding a currency to the book does not make the simulation slower; it makes the number of simulations larger:

Curve points	Vol points	Bump-and-revalue	AAD
50	30	81 runs	~4 runs
80	45	126 runs	~4 runs
120	70	191 runs	~4 runs

At twelve minutes per pricing run, eighty-one runs is **16.2 hours** and does not fit in an overnight window; four is **48 minutes**. The right column does not move, which is the whole of the argument for adjoint methods.

Where the payoff permits it, Section 20.10's pathwise differentiation beats bumping: differentiating the simulated payoff with respect to the input along each path gives an unbiased estimator out of a single simulation, with no bump size to choose. It fails for discontinuous

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payoffs (a digital, or a barrier), where the payoff has no derivative to take, and the repairs are to smooth the payoff or to fall back on the likelihood-ratio method.

Adjoint algorithmic differentiation changed the economics of the whole exercise. AAD (Section 20.10) computes all $M + V$ sensitivities in a fixed multiple (typically three to five times) of the cost of one price, independently of how many inputs there are. Eighty-one simulations become four, which is a factor of twenty and is the reason models of this complexity are viable on a trading desk.

Section 9.1 noted that cost, not method, is the binding constraint on an XVA desk's exposure calculation; the same logic operates here. A model that cannot produce its own risk overnight will not be used, however correct it is, and AAD is what moved several models across that line after 2010.

32.14 PRODUCTS THAT REQUIRE IT

The abstract case for more factors is Section 31.12. Here is the concrete list, because the model exists for these and for very little else.

CMS spread options and the steepener notes built on them. A coupon paying $\max(S_{10} - S_2 - K, 0)$, sold to investors who want yield and will be long the curve's slope to get it: the coupon pays only while the ten-year rate exceeds the two-year by more than K, which a merely flat curve already fails, let alone an inverted one. Example 31.11 measured what one factor costs here. These are the largest single use of the model.

Range accruals on two rates. The coupon accrues only on days when a rate (or the spread between two) sits inside a band. The payoff depends on the joint distribution at many dates, not on one date's marginal.

Ratchets and target redemption notes. A ratchet's coupon is capped relative to the previous coupon, which makes it path-dependent in the forward-rate vector; a **target redemption note (TARN)** terminates once cumulative coupons reach a threshold, which makes the termination date itself a functional of the path. Neither is a function of a single state variable at a single date, which is precisely the condition under which Chapter 31's tree stops being applicable.

Callable structured notes, which combine the above with the exercise problem of Section 32.12, and are the hardest products in the rate market for exactly that reason.

Who is on each side is worth stating, because it explains the market's shape. The buyers are yield-seeking investors: private banks, insurers, regional banks. In a callable note or a range accrual they are short optionality and are paid a coupon for it, but the steepener's buyer is long a CMS spread option, and the steepener is what sets the correlation market. The dealer is therefore short spread optionality and must hedge it, which means trading swaptions and CMS and carrying the correlation risk of Section 32.8. A spread option is worth more the less the two rates are correlated, so the large and persistent one-way demand that structured-note issuance creates bids spread volatility up, and that is why dealers' implied correlations have tended to sit below historical ones. It is the same asymmetry Section 19.10 identified in equity volatility, arising the same way.

32.15 SMILE

The LMM as stated is lognormal, so it produces no smile, and Section 29.9 established that rate markets have a pronounced one. Two repairs are in use.

The shifted or displaced LMM models $F_k + s$ as lognormal, which produces a skew and, by the same mechanism as Section 31.14's shifted model, accommodates negative rates. It is cheap and is what most books use for the skew alone; a book quoted in normal volatility uses it or the normal LMM.

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SABR-LMM gives each forward its own stochastic volatility with the SABR dynamics of Section 29.9, correlated across forwards. This is the state of the art and it is expensive: the parameter count multiplies, and the correlation structure now has to cover volatility-volatility and rate-volatility correlations as well as rate-rate.

The market's actual practice is worth recording because it is inconsistent and works. Swaptions are quoted through SABR per expiration-tenor cell (the volatility cube of Section 29.9), and the LMM is calibrated to those quotes. So SABR does the smile interpolation and the LMM does the term-structure dynamics, and the two are not a single consistent model. Section 21.12's judgment applies: a model used for interpolation and a model used for dynamics are being asked different questions, and requiring one object to do both has costs that are usually larger than the inconsistency.

32.16 MULTI-CURVE, AFTER THE TRANSITION

Chapter 7 rebuilt the framework around a discount curve and one or more forecast curves. What does that do to a model whose state variable is a forward rate?

Not much, as with the short-rate models of Chapter 31, but for the opposite reason. A short-rate model models one risk-free rate, and any spread is added to it as a deterministic amount. An LMM models whatever forward rates you tell it to, so a multi-curve LMM simply carries more of them (the forwards of each index's forecast curve, plus the discount curve), with the drift condition applying to each set under the chosen numeraire. Two genuine changes follow.

The numeraire is now the discount curve's bond, so the drift (32.7) is taken with respect to the discounting curve while the forwards being modeled come from forecast curves. That separation is precisely what (32.2) ruled out: a forward curve and a discount curve carrying the same information, either determining the other. The identity still holds within each curve, but it no longer runs between them, so the two must be carried and calibrated separately. The drift correction this requires is the covariance of the basis with the discount forwards, and most implementations set it to zero by treating that basis as deterministic. That is reasonable while the basis is small and stable, which the move to overnight rates largely made it (Section 33.9 traces the tenor basis shrinking to term premia and compounding mechanics once the credit component went), and it is wrong for anything whose payoff depends on the basis itself, such as the currency option inside a two-currency CSA that Section 9.6 prices as ColVA.

Backward-looking rates also change what a forward is. Section 29.4's compounded-in-arrears rate is not known at the start of its period, so the "forward rate" for a SOFR period does not fix at the start of its accrual period: it keeps moving, as a martingale, until the period ends. The clean formulations of this (the generalized forward market model) were developed after 2018, and they show that the received LMM machinery survives with the accrual convention changed. The transition altered what is compounded, not how the model is built, which is a fair summary of its effect on this whole chapter.

32.17 CHOOSING A MODEL

The reader who skips the rest of the chapter should take this section.

Use a one-factor short-rate model when the payoff depends on the level of rates and the product has early exercise: callable bonds, Bermudan swaptions, mortgage books. Chapter 31's speed and natural treatment of exercise decide it, and Example 31.9's calibration warning is the price.

Use a two-factor short-rate model when the payoff has mild shape dependence and still needs exercise. It keeps the lattice, at the cost of a two-dimensional one.

Use an LMM when the payoff depends on several rates at once and on their joint distribution:

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CMS spread options, ratchets, range accruals on more than one rate, anything where Section 32.8's correlation matrix is the answer to the pricing question. Accept that early exercise will cost you regression and time.

Use SABR alone when the question is interpolation across strikes at one expiration, which is most of what a swaption desk does most days.

The general principle outlasts all four: a model should be chosen by which of its errors you can afford, not by which is most realistic. Chapter 31's one-factor model is unrealistic in a way that costs a factor of four on a spread option and nothing, through correlation, on a callable bond. The LMM is realistic in a way that costs seconds instead of milliseconds. Neither fact is a defect; both are the specification.

32.18 WHO USES THESE

Exotic rate desks at the large dealers, which is where the LMM lives. The products it exists for (CMS spread options, structured coupons on more than one rate) are dealer-intermediated and issued into structured notes, so the model's user and the product's ultimate holder are several steps apart.

Model validation groups, disproportionately. An LMM implementation has more places to be wrong than any other model in this book (drift, measure, factor reduction, regression basis, discretization), and the validation effort is correspondingly large.

Insurers with variable annuity books (investment-linked savings policies that carry a guaranteed minimum benefit), where the guarantee depends jointly on equity and rate paths and a consistent multi-asset simulation is unavoidable.

And academics and vendors, more than trading desks. This should be said plainly: the LMM is the most-written-about interest rate model and not the most-used one. Chapter 31's models price more notional every day. The LMM's importance is that it defines what the simpler models are approximating, and Example 32.5 is an instance—you cannot know that a structural inconsistency is worth about a basis point without a model in which it is absent.

32.19 WHERE THIS CAME FROM

Heath, Jarrow, and Morton (1992) generalized Chapter 31's models by dropping the short rate altogether: model the whole forward curve, and derive the condition its drift must satisfy for no arbitrage. HJM is a framework rather than a model, and its practical difficulties are the two Section 32.3 identifies: the process is generally non-Markov, and no instrument is quoted on an instantaneous forward rate, so the volatility being modeled is not one any quote reveals.

Brace, Gatarek, and Musiela (1997), Jamshidian (1997), and Miltersen, Sandmann, and Sondermann (1997) fixed the second within a year of each other by modeling the rates the market actually quotes, and accepted the first: a market model is simulated, not put on a tree. The result was not only a usable model but a justification: it showed that the Black formula the market had been applying to caps, and separately to swaptions, for years was consistent with an arbitrage-free term structure, which until then no one had established.

SUMMARY

HJM models every instantaneous forward rate at once, with today's curve as the initial condition, so Chapter 31's fit to the curve is built in rather than solved. Once the volatility functions are chosen, no-arbitrage determines the drift, and every arbitrage-free term-structure model is an HJM model. The framework is not implemented directly because its input is a volatility for every instantaneous forward, which no quote reveals, and because the process is generally non-Markov and so has no recombining tree; Hull–White's exponential volatility, with Ho–Lee's

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flat one as its limit, is the simplest structure that keeps it Markov, which is why Chapter 31 could use a lattice at all.

The LIBOR market model fixes a tenor structure and makes each forward lognormal, so Black's caplet formula is reproduced by construction rather than by calibration. The number of factors it needs depends on the correlation structure a product is written on: one factor leaves much of that variance unexplained. The often-quoted 99% comes from principal components of yield changes, and yields, which average the forwards, move together far more than forwards do, so it does not describe the forward-rate correlations a spread option depends on.

The market fixes each forward's total variance and not the shape of its instantaneous volatility, so Rebonato's humped time-homogeneous form is chosen for stability rather than fit. Terminal and instantaneous correlation differ by little, so the correlation you calibrate is nearly the correlation you get (though a spread's variance amplifies even that small gap), and Chapter 31's one-factor defect needs more factors rather than cleverer volatility.

In simulation, the predictor–corrector is worth its cost only when the drift is large. Forwards and swap rates cannot both be exactly lognormal, so the market's two quoting conventions are inconsistent, but at the money the inconsistency is far inside the bid–ask: a theorem with no practical cost. Smile enters through a shifted or SABR-based LMM, and in market practice SABR interpolates across strikes while the LMM supplies the dynamics. The move to overnight rates changed what is compounded, not how the model is built.

Sensitivities, not prices, are the binding constraint: bump-and-revalue costs a simulation per input, while adjoint algorithmic differentiation delivers all of them at a small multiple of the cost of one price. The products that require the model (CMS spread options, two-rate range accruals, ratchets, and TARNs) depend on several rates at several dates, and dealers' one-way issuance flow is why their implied correlations have tended to sit below historical ones. Early exercise costs the LMM its speed, which is why short-rate models still price the callable book. Choose a model by which of its errors you can afford, not by which is most realistic.

FURTHER READING

Heath, Jarrow, and Morton's paper is the origin of Section 32.2 and the original statement of why the drift is not free; it is harder going than the market-model papers and worth the effort once. Brace, Gatarek, and Musiela, and independently Jamshidian and Miltersen, Sandmann, and Sondermann, established the market model and its consistency with Black's formula, and the original papers are the place to see the historical point of Section 32.4 as it was first made. Litterman and Scheinkman is the source of the principal-component reading of Example 32.1, and is where level, slope and curvature were first named as the factors of a curve; it is short. Rebonato's books are the practitioner's reference on everything in Sections 32.7 through 32.9, he treats the volatility-correlation trade-off in calibration at length, and the parametric correlation forms of Example 32.1, including the long-end floor, are his. Hull and White take the other route through the same material: the piecewise-constant instantaneous volatility of Section 32.7, bootstrapped period by period off the caplet Black volatilities, is theirs, and the same paper relates forward-rate to swap-rate volatilities by an approximation of the same kind as the one Section 32.11 measures. Brigo and Mercurio cover the algebra of the drift conditions and stochastic-volatility extensions of the LMM. Andersen and Piterbarg are the reference for simulation and for the early-exercise machinery of Section 32.12, including the dual upper bounds this chapter only mentions. On backward-looking rates and the post-transition formulations of Section 32.16, the literature is recent and still settling; the paper of Lyashenko and Mercurio is the standard entry point.

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PRACTICE QUESTIONS

32.1 Explain what the state variable is in an HJM model, why today's curve poses no fitting problem, and state the HJM drift condition in words together with what it says about the model's degrees of freedom.

32.2 Explain in what sense Ho–Lee and Hull–White are HJM models, why a general HJM model cannot be put on a recombining tree, and identify the volatility structure that makes one Markov.

32.3 Explain why the LIBOR market model reproduces Black's caplet formula by construction rather than by calibration, and why the forward rates cannot all be martingales under one measure.

32.4 Compute $\tau F/(1+\tau F)$ at $F = 3\%$ and at $F = 8\%$ with annual accrual, and explain the relevance to the drift.

32.5 Using Example 32.1, state the share of variance explained by one and by three factors at $\beta = 0.05$, and explain why the often-quoted figure of 99% for three factors is misleading in this context.

32.6 Distinguish flat, spot, and instantaneous volatility, state which the market quotes, and explain why the market determines the integral of instantaneous variance but not its shape.

32.7 Explain why a time-homogeneous volatility parametrization is preferred, and relate this to Section 21.3.

32.8 Distinguish instantaneous from terminal correlation, state which one a spread option pays on, use Example 32.3 to give the maximum decorrelation produced by volatility shape alone, and explain why that result strengthens rather than weakens Chapter 31's criticism.

32.9 Describe the three stages of LMM calibration, state which is exact and which is underdetermined, and explain what is lost when a full correlation matrix is reduced to K factors.

32.10 Using Example 32.4, state the change in implied volatility between 4 and 200 time steps, explain why the predictor–corrector makes no difference there, and state the conditions under which it would.

32.11 Explain why forward rates and swap rates cannot both be exactly lognormal, explain the significance of Rebonato's weights summing to one, and use Example 32.5 to state the size of the inconsistency in each of the two cases and compare it with a typical bid–ask.

32.12 Compute the number of simulations a bump-and-revalue risk run requires for a book with forty curve points and twenty-five volatility points, and explain why common random numbers are essential rather than merely helpful.

32.13 State the condition under which pathwise differentiation fails and name one repair, then explain what adjoint algorithmic differentiation delivers and why its cost does not grow with the number of inputs.

32.14 Name four products that require a multi-factor model, state for each what makes a single state variable insufficient, and explain why dealers' implied correlations have tended to sit below historical ones.

32.15 Explain why using the same Monte Carlo paths to fit and to apply an exercise rule biases the price upward, and why a suboptimal rule produces a lower bound rather than an error of unknown sign.

32.16 State which model you would use for a callable bond, for a CMS spread option, and for

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quoting a single swaption, with reasons.

FURTHER QUESTIONS

32.17 Derive the HJM drift condition for a one-factor model by imposing that discounted bond prices are martingales. Then verify that $s(t,T) = \sigma e^{-a(T-t)}$ reduces the resulting dynamics to a Markov process in a single state variable, and identify that variable.

32.18 Example 32.1 shows three factors explaining 78–89% of variance on an assumed correlation structure, against the often-quoted 99% on yield changes. Reconcile the two figures precisely, determine which is relevant for a CMS spread option and which for a duration hedge, and state the general principle for deciding which decomposition a given problem needs.

32.19 Example 32.3 finds the volatility-shape channel contributes at most 0.6% of decorrelation under Rebonato's parametrization. Determine whether this is a property of the parametrization or a general feature: construct a volatility structure that maximizes the gap between instantaneous and terminal correlation, state how large it can be made, and judge whether such a structure is economically plausible.

32.20 Example 32.4 shows discretization bias dominating drift-correction error. Identify the source of the residual bias, propose a scheme that would remove it, and estimate the number of time steps at which the remaining bias falls below the Monte Carlo standard error for a book of one million paths.

32.21 Example 32.5 finds the swap/forward incompatibility worth at most about a basis point of volatility on both a short and a long trade, although the weights w_k that Rebonato's approximation freezes demonstrably move over the option's life. Derive the approximation's error term explicitly, show that at the money it is second order rather than first order in the weights' displacements, and explain what the constraint $\sum_k w_k = 1$ has to do with it.

32.22 Section 32.15 notes that the market lets SABR do smile interpolation and the LMM do term-structure dynamics, which is inconsistent. Construct the argument that this should be replaced by a single consistent model, construct the argument that it should not, and state what evidence would settle it.

32.23 Section 32.17 argues a model should be chosen by which of its errors you can afford. Apply this to a specific book (a portfolio of Bermudan swaptions and CMS spread options in one currency) and design the model configuration you would run, stating explicitly which errors you are accepting on which trades and how you would monitor them.

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